Arithmétique 01

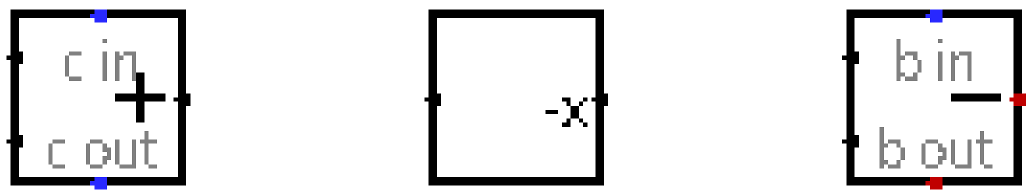


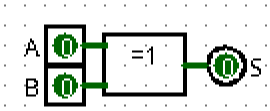
Table des matières

1) Additionneur 1 bit :

1-1) Sans la retenue :

Entrée		Sortie
A	B	S
0	0	0
0	1	1
1	0	1
1	1	0

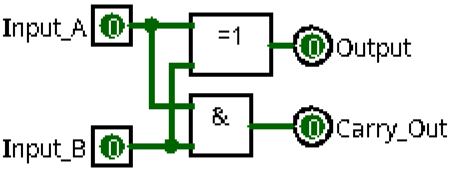
A	0	0	1	1
+ B	+ 0	+ 1	+ 0	+ 1
= S	= 0	= 1	= 1	= 0

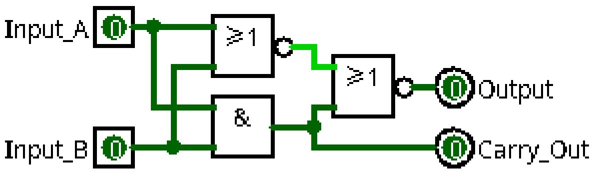


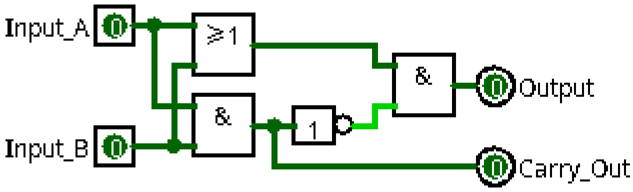
1-2) Avec la retenue de sortie : (ADDSP0 / ADDSP1 / ADDSP2)

A	0	0	1	1
+ B	+ 0	+ 1	+ 0	+ 1
= S	= 0	= 1	= 1	= 0
RS	0	0	0	1

Entrée		Sortie	
A	B	RS	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0







1-3 Avec la retenue : (ADD0 / ADD1 / ADD2)

	R	0	0	0	0
+	A	0	0	1	1
+	B	0	1	0	1
=	S	0	1	1	0
	RS	0	0	0	1

	R	1	1	1	1
+	A	0	0	1	1
+	B	0	1	0	1
=	S	1	0	0	1
	RS	0	1	1	1

Entrée			Sortie	
R	A	B	RS	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

R	A	B	S	S=
0	0	0	0	0
0	0	1	1	$\overline{R}\cdot\overline{A}\cdot B$
0	1	0	1	$\overline{R}\cdot A\cdot\overline{B}$
0	1	1	0	0
1	0	0	1	$R\cdot\overline{A}\cdot\overline{B}$
1	0	1	0	0
1	1	0	0	0
1	1	1	1	$R\cdot A\cdot B$

R	A	B	RS	RS=
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	1	$\overline{R}\cdot A\cdot B$
1	0	0	0	0
1	0	1	1	$R\cdot\overline{A}\cdot B$
1	1	0	1	$R\cdot A\cdot\overline{B}$
1	1	1	1	$R\cdot A\cdot B$

Algèbre de Boole

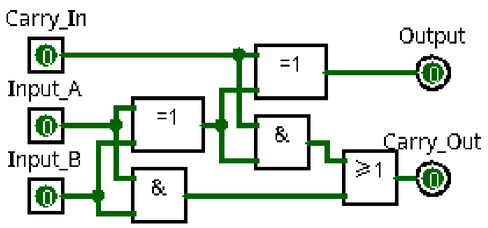
$$S=(\overline{R}\cdot\overline{A}\cdot B)+(\overline{R}\cdot A\cdot\overline{B})+(R\cdot\overline{A}\cdot\overline{B})+(R\cdot A\cdot B)$$

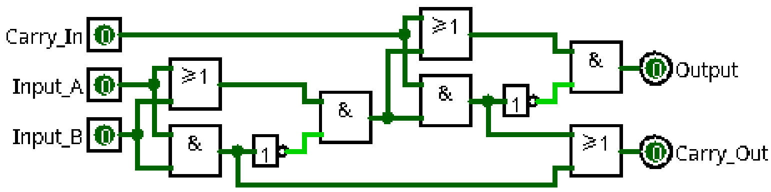
$$S=(\overline{R}\cdot\overline{A}\cdot B)+(\overline{R}\cdot A\cdot\overline{B})+(R\cdot\overline{A}\cdot\overline{B})+(R\cdot A\cdot B)$$
$$S=\overline{R}\cdot\overline{A}\cdot B+\overline{R}\cdot A\cdot\overline{B}+R\cdot\overline{A}\cdot\overline{B}+R\cdot A\cdot B$$
$$S=\overline{R}\cdot(\overline{A}\cdot B+A\cdot\overline{B})+R\cdot(\overline{A}\cdot\overline{B}+A\cdot B)$$
$$S=\overline{R}\cdot(A\oplus B)+R\cdot(\overline{A\oplus B})$$
$$S=R\oplus(A\oplus B)$$
$$S=(A\oplus B)\oplus R$$

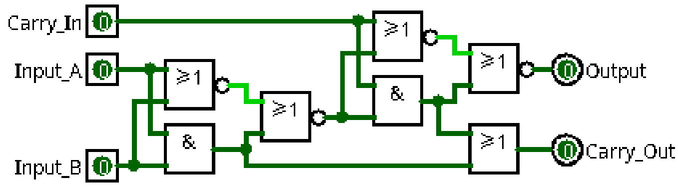
Algèbre de Boole

$$RS=(\overline{R}\cdot A\cdot B)+(R\cdot\overline{A}\cdot B)+(R\cdot A\cdot\overline{B})+(R\cdot A\cdot B)$$

$$RS=(\overline{R}\cdot A\cdot B)+(R\cdot\overline{A}\cdot B)+(R\cdot A\cdot\overline{B})+(R\cdot A\cdot B)$$
$$RS=\overline{R}\cdot A\cdot B+R\cdot\overline{A}\cdot B+R\cdot A\cdot\overline{B}+R\cdot A\cdot B$$
$$RS=\overline{R}\cdot A\cdot B+R\cdot A\cdot B+R\cdot\overline{A}\cdot B+R\cdot A\cdot\overline{B}$$
$$RS=(\overline{R}+R)\cdot(A\cdot B)+R\cdot(\overline{A}\cdot B+A\cdot\overline{B})$$
$$RS=1\cdot(A\cdot B)+R\cdot(A\oplus B)$$
$$RS=(A\cdot B)+((A\oplus B)\cdot R)$$



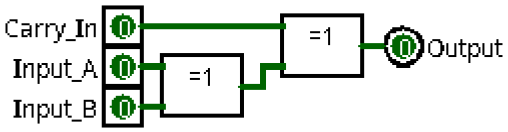


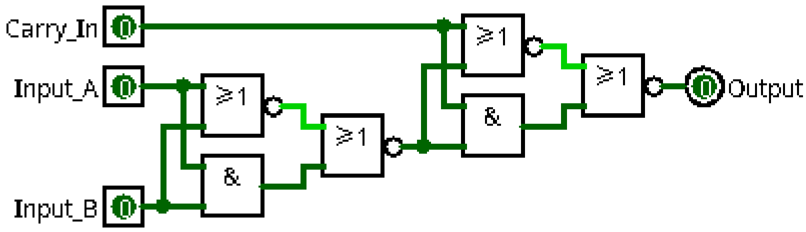


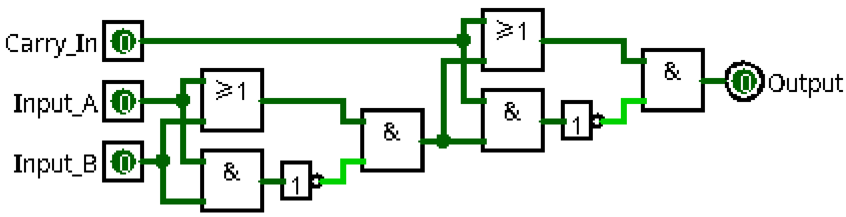
1-4) Sans la retenue de sortie : (ADDSR0 / ADDSR1 / ADDSR2)

R	0	0	0	0
+ A	+ 0	+ 0	+ 1	+ 1
+ B	+ 0	+ 1	+ 0	+ 1
= S	= 0	= 1	= 1	= 0

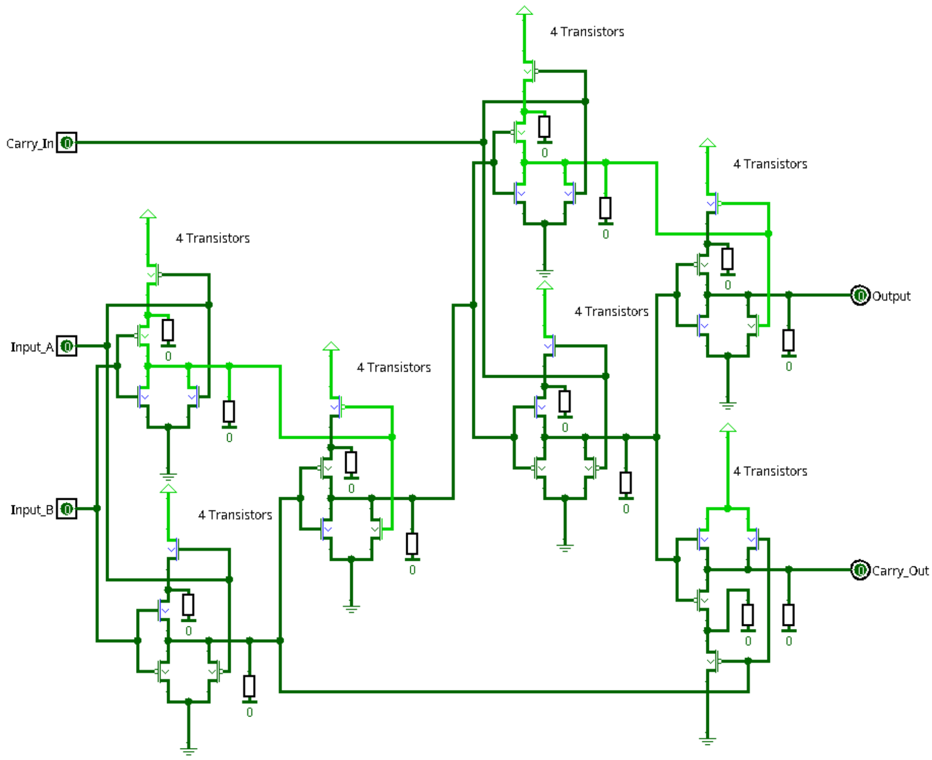
R	1	1	1	1
+ A	+ 0	+ 0	+ 1	+ 1
+ B	+ 0	+ 1	+ 0	+ 1
= S	= 1	= 0	= 0	= 1





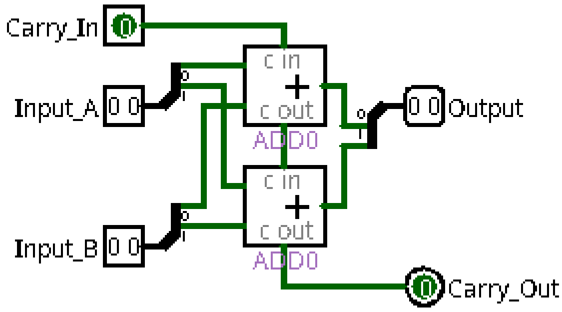


1-5) Transistors : (ADDTR)

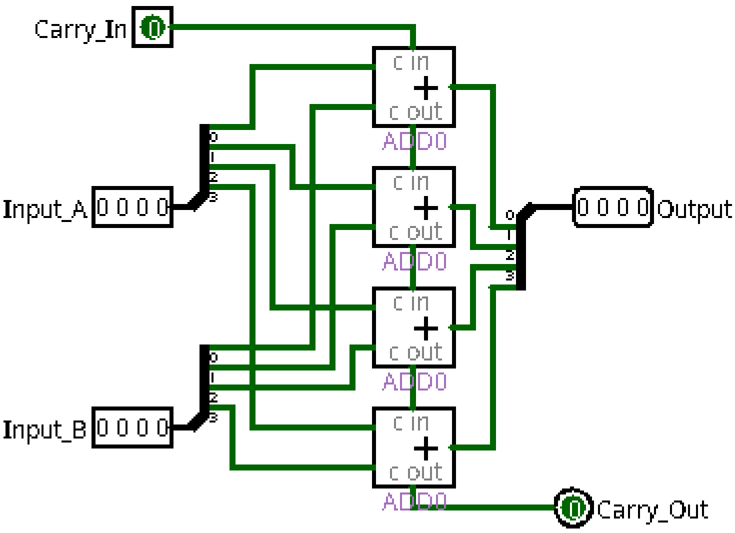


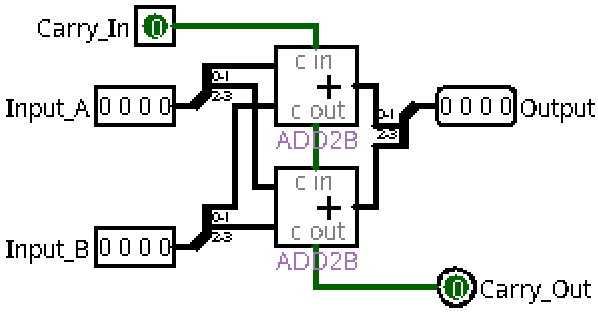
2) Additionneur :

2-1) 2 bits :

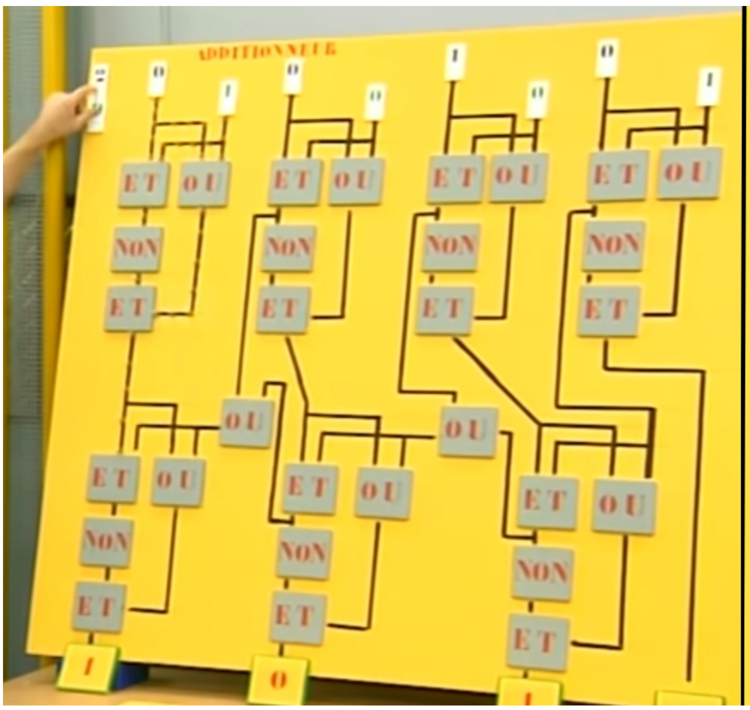


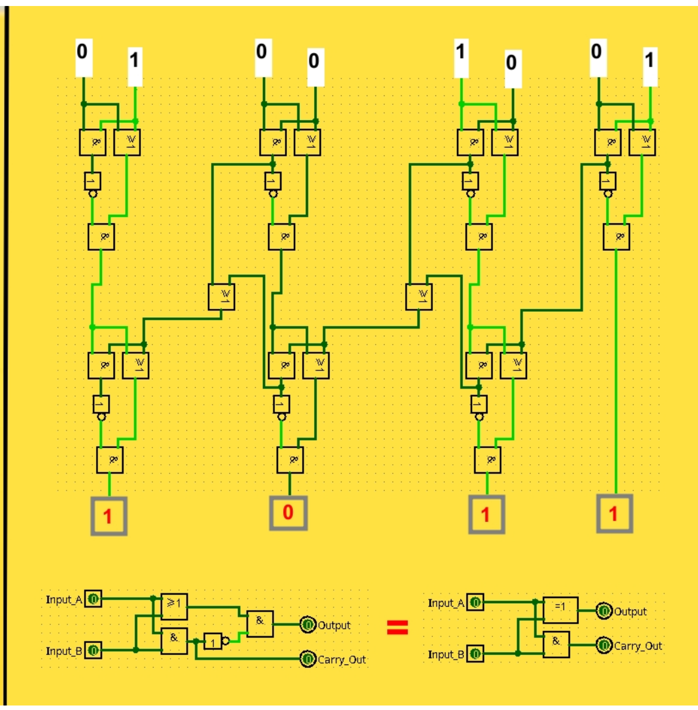
2-2) 4 bits :

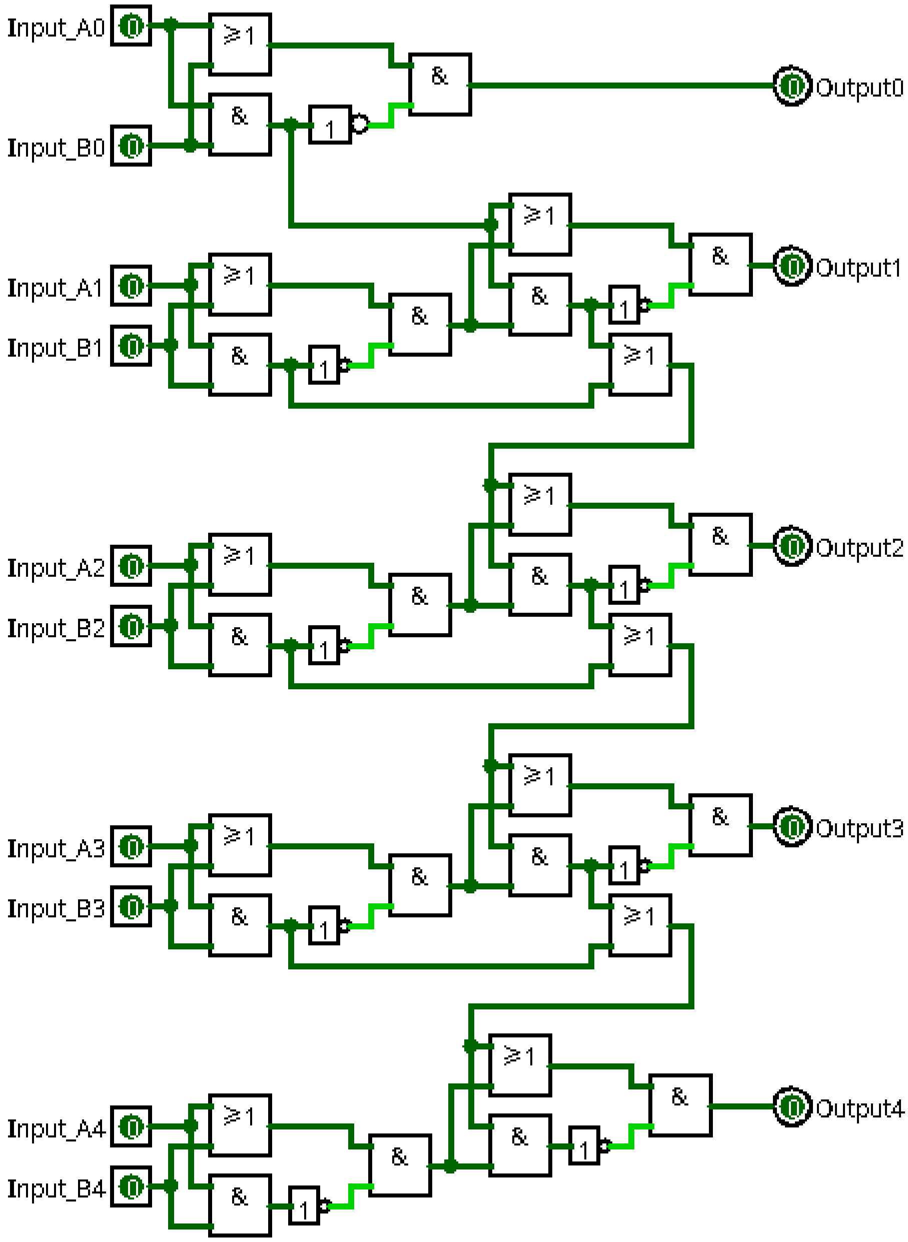


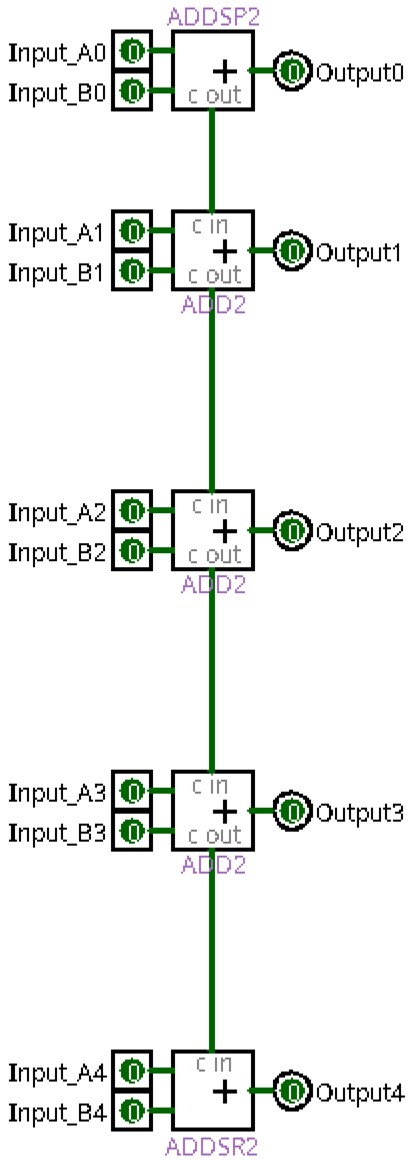


2-3) C'est pas sorcier :






Input_A0
Input_B0
Output0
Input_A1
Input_B1
Output1
Input_A2
Input_B2
Output2
Input_A3
Input_B3
Output3
Input_A4
Input_B4
Output4


ADDSP2
Input_A0
Input_B0
Output0
c out
Input_A1
Input_B1
c in
Output1
ADD2
Input_A2
Input_B2
Output2
ADD2
Input_A3
Input_B3
Output3
ADD2
Input_A4
Input_B4
Output4
ADDSR2

3) Nombre négatif :

3-1) Binaire :

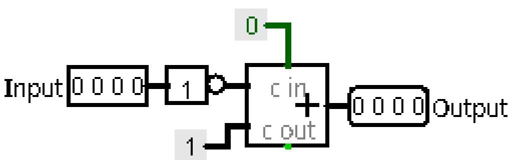
Binaire				Décimal
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	-8
1	0	0	1	-7
1	0	1	0	-6
1	0	1	1	-5
1	1	0	0	-4
1	1	0	1	-3
1	1	1	0	-2
1	1	1	1	-1

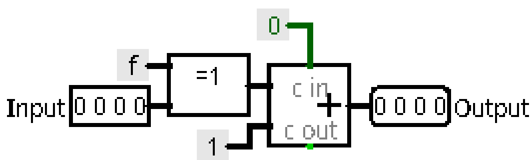
	0	0	0	1	1
NOT	1	1	1	0	-2
+1	1	1	1	1	-1
	1	1	1	1	-1
NOT	0	0	0	0	0
+1	0	0	0	1	1
	0	1	1	0	6
NOT	1	0	0	1	0
+1	1	0	1	0	-6
	1	0	1	0	-6
NOT	0	1	0	1	5
+1	0	1	1	0	6

3-2) Ne pas faire :

	0	0	0	1	1
+1	0	0	1	0	2
NOT	1	1	0	1	-3
	1	1	1	1	-1
+1	0	0	0	0	0
NOT	0	0	0	1	1
	0	1	1	0	6
+1	0	1	1	1	7
NOT	1	0	0	0	-8
	1	0	1	0	-6
+1	1	0	1	1	-5
NOT	0	1	0	0	4

3-3) Inverseur de signe :



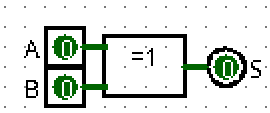


4) Soustraction 1 bit :

4-1) Sans la retenue :

Entrée		Sortie
A	B	S
0	0	0
0	1	1
1	0	1
1	1	0

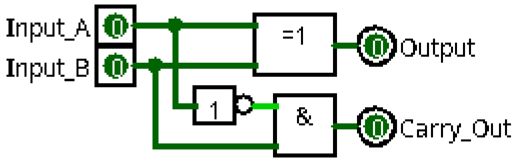
A	0	0	1	1
- B	- 0	- 1	- 0	- 1
= S	= 0	= 1	= 1	= 0

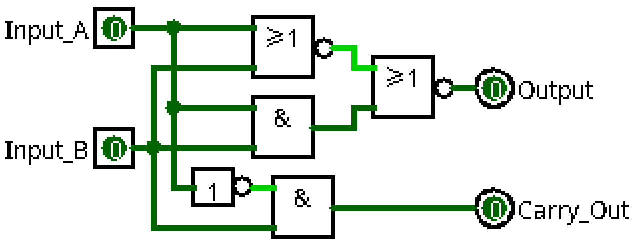


4-2) Avec la retenue de sortie : (SUBSP0 / SUBSP1)

A	0	0	1	1
- B	- 0	- 1	- 0	- 1
= S	= 0	= 1	= 1	= 0
RS	0	1	0	0

Entrée		Sortie	
A	B	RS	S
0	0	0	0
0	1	1	1
1	0	0	1
1	1	0	0





4-3 Avec la retenue : (SUB0 / SUB1)

A	0	0	1	1
- (R+B)	- (0+0)	- (0+1)	- (0+0)	- (0+1)
A	0	0	1	1
- (R+B)	- 0	- 1	- 0	- 1
= S	= 0	= 1	= 1	= 0
RS	0	1	0	0

A	0	0	1	1
- (R+B)	- (1+0)	- (1+1)	- (1+0)	- (1+1)
A	0	0	1	1
- (R+B)	- 1	- 0	- 1	- 0
= S	= 1	= 0	= 0	= 1
RS	1	1	0	1

Entrée			Sortie	
R	A	B	RS	S
0	0	0	0	0
0	0	1	1	1
0	1	0	0	1
0	1	1	0	0
1	0	0	1	1
1	0	1	1	0
1	1	0	0	0
1	1	1	1	1

Entrée			Sortie	
R	A	B	S	S=
0	0	0	0	0
0	0	1	1	$\overline{R}\cdot\overline{A}\cdot B$
0	1	0	1	$\overline{R}\cdot A\cdot\overline{B}$
0	1	1	0	0
1	0	0	1	$R\cdot\overline{A}\cdot\overline{B}$
1	0	1	0	0
1	1	0	0	0
1	1	1	1	$R\cdot A\cdot B$

Entrée			Sortie	
R	A	B	RS	RS=
0	0	0	0	0
0	0	1	1	$\overline{R}\cdot\overline{A}\cdot B$
0	1	0	0	0
0	1	1	0	0
1	0	0	1	$R\cdot\overline{A}\cdot\overline{B}$
1	0	1	1	$R\cdot\overline{A}\cdot B$
1	1	0	0	0
1	1	1	1	$R\cdot A\cdot B$

Algèbre de Boole

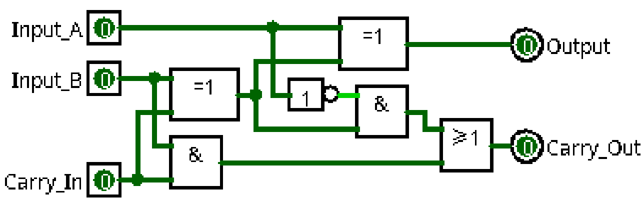
$$S=\overline{A}\cdot\overline{R}\cdot B+\overline{A}\cdot R\cdot\overline{B}+A\cdot\overline{R}\cdot\overline{B}+A\cdot R\cdot B$$

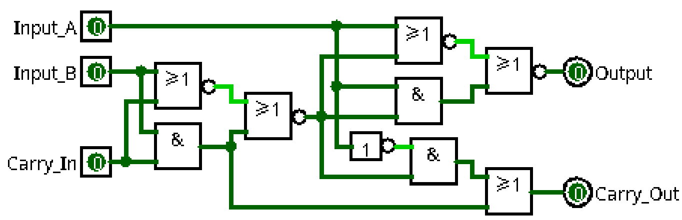
$$S=\overline{A}\cdot\overline{R}\cdot B+\overline{A}\cdot R\cdot\overline{B}+A\cdot\overline{R}\cdot\overline{B}+A\cdot R\cdot B$$
$$S=\overline{A}\cdot(\overline{R}\cdot B+R\cdot\overline{B})+A\cdot(\overline{R}\cdot\overline{B}+R\cdot B)$$
$$S=\overline{A}\cdot(R\oplus B)+A\cdot(\overline{R\oplus B})$$
$$S=A\oplus(R\oplus B)$$
$$S=(R\oplus B)\oplus A$$

Algèbre de Boole

$$RS=\overline{A}\cdot\overline{R}\cdot B+\overline{A}\cdot R\cdot\overline{B}+\overline{A}\cdot R\cdot B+A\cdot R\cdot B$$

$$RS=\overline{A}\cdot\overline{R}\cdot B+\overline{A}\cdot R\cdot\overline{B}+\overline{A}\cdot R\cdot B+A\cdot R\cdot B$$
$$RS=\overline{A}\cdot(\overline{R}\cdot B+R\cdot\overline{B})+(\overline{A}+A)\cdot(R\cdot B)$$
$$RS=\overline{A}\cdot(B\oplus R)+1\cdot(R\cdot B)$$
$$RS=\overline{A}\cdot(B\oplus R)+(R\cdot B)$$
$$RS=(R\cdot B)+((B\oplus R)\cdot\overline{A})$$

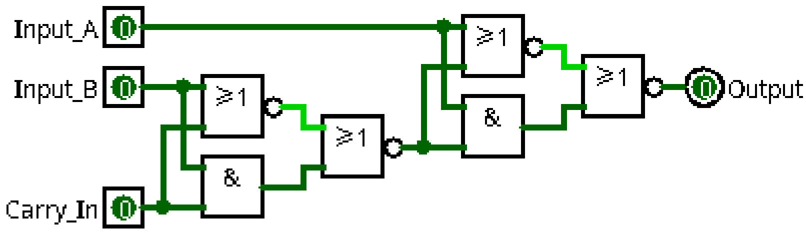


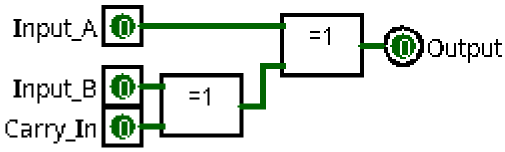


4-4) Sans la retenue de sortie : (SUBSR0 / SUBSR12)

A	0	0	1	1
- (R+B)	- (0+0)	- (0+1)	- (0+0)	- (0+1)
A	0	0	1	1
- (R+B)	- 0	- 1	- 0	- 1
= S	= 0	= 1	= 1	= 0

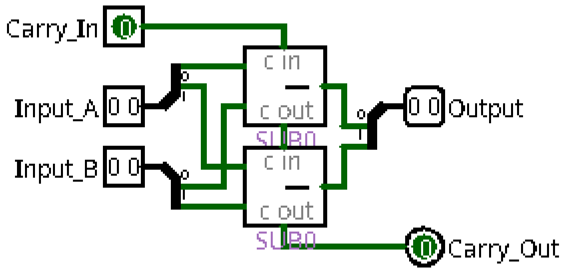
A	0	0	1	1
- (R+B)	- (1+0)	- (1+1)	- (1+0)	- (1+1)
A	0	0	1	1
- (R+B)	- 1	- 0	- 1	- 0
= S	= 1	= 0	= 0	= 1



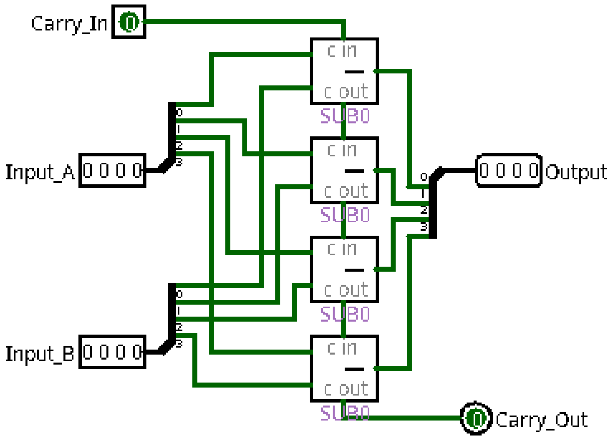


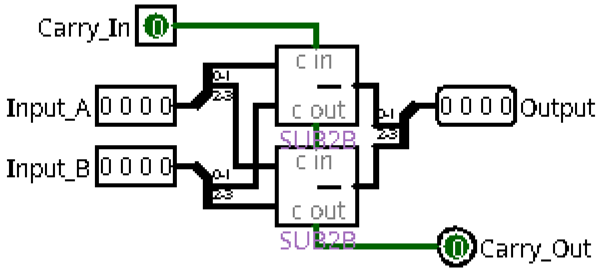
5) Soustraction :

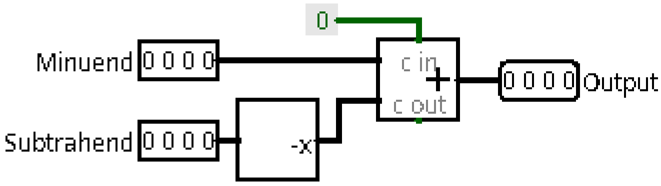
5-1) 2 bits :



5-2) 4 bits :


Carry_In
0
Input_A
0 0 0 0
Input_B
0 0 0 0
c in
c out
SUB0
0 0 0 0
Output
0
Carry_Out


Carry_In
0
Input_A
0 0 0 0
Input_B
0 0 0 0
0-1
2-3
c in
c out
SUB2B
0 0 0 0
Output
0
Carry_Out


0
Minuend
0 0 0 0
Subtrahend
0 0 0 0
-x
c in
+
c out
0 0 0 0
Output